Main Battery System – Maintenance Practices

1. Removal/Installation

A. Remove Main Battery

CAUTION **BEFORE DISCONNECTING THE BATTERY, ENSURE THAT THE MASTER SWITCH (PRE MOD 6/2314) OR BATTERY SWITCH (POST MOD 6/2314) IS OFF, AND PULL COMP LTS CIRCUIT BREAKER. ANY SUBSTANTIAL LOAD ON THE BATTERY COULD CAUSE DAMAGE TO CONNECTOR PINS AND SOCKETS WHEN BATTERY IS BEING DISCONNECTED.**

Before removing the battery, refer to safety precautions in 24-00-00, Electrical Power – General Data. Refer to Figure 1 and Figure 2 for details.

1) Set the MASTER switch to OFF (Pre Mod 6/2314); or set BATTERY and EXTERNAL switches to OFF (Post Mod 6/2314).

2) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3) Remove the access panel in the rear of the baggage compartment floor. For aircraft with S.O.O. 6371, gain access to the battery through the nose compartment door. (Refer to 12-00-00, Servicing – General Data, Figure 2 – Servicing Diagram)

Note If S.O.O. 6431 is incorporated, remove upper and lower bulkhead panels, removing the bolts which attach the bulkhead panels to the securing brackets.

4) Disconnect the battery quick disconnect cable.

5) Remove safety wire and battery tie down bars.

6) Disengage the battery.

WARNING **BATTERIES ARE VERY HEAVY. USE APPROPRIATE LIFTING DEVICES OR EQUIPMENT. USE BATTERY HANDLES WHERE PROVIDED.**

7) Remove the battery from the tray. For aircraft with S.O.O. 6371, ensure to protect nose composite floor with hard foam or plywood to prevent impact damage due to the battery dropping onto the composite structure.

Note Ideally, hard foam or plywood would sit at the same height as the surface which the battery rests on, allowing the battery to be slid off the tray (preventing the need to lift the battery off the mounting tray).

B. Install Main Battery

1) Inspect the battery for damage. Cracks in metal or plastic containers are not permitted. Dents in the metal containers that impinge on the interior plastic container are not acceptable.

2) Set the MASTER switch to OFF (Pre Mod 6/2314); or set BATTERY and EXTERNAL switches to OFF (Post Mod 6/2314).

3) Disconnect DC Power (refer to 12-00-00, Servicing – General Data, Para 2.).

4) Remove the access panel in the rear of the baggage compartment floor. For aircraft with S.O.O. 6371 incorporated, place hard foam or plywood beside mounting tray, ensuring it is at the same level as the battery tray. Gently place the battery on the hard foam or plywood and slide battery into place.

Note If S.O.O. 6431 is incorporated, remove upper and lower bulkhead panels, removing the bolts which attach the bulkhead panels to the securing brackets.

5) Ensure the battery tray is clean and dry.

WARNING **BATTERIES ARE VERY HEAVY. USE APPROPRIATE LIFTING DEVICES OR EQUIPMENT. USE BATTERY HANDLES WHERE PROVIDED.**

6) Install the battery in the container.

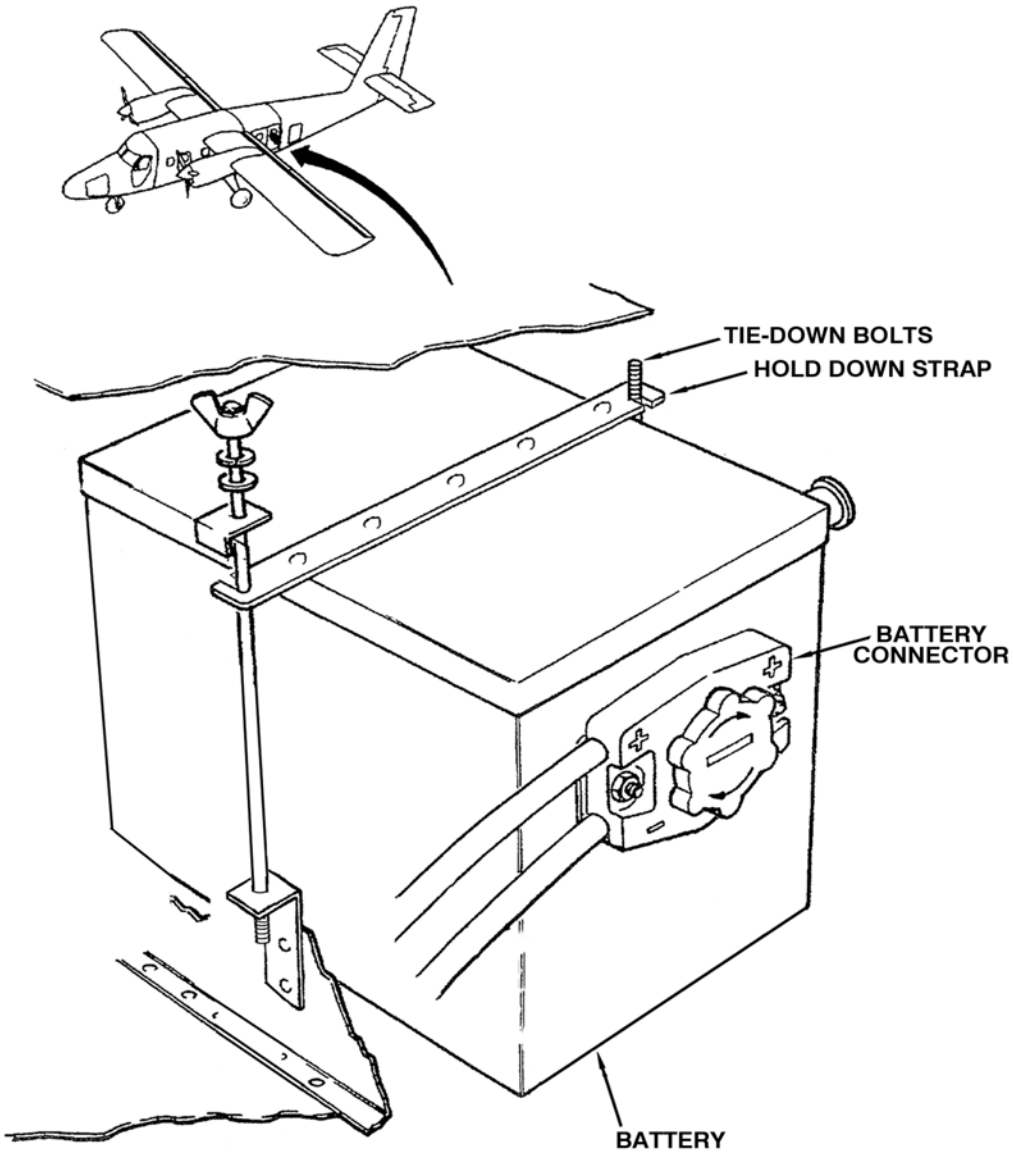
CAUTION **USE AN OPEN END WRENCH ON THE FLATS OF THE BATTERY TERMINAL, WHERE AVAILABLE, WHILE TORQUING TERMINAL BOLTS. FAILURE TO DO SO MAY RESULT IN RUPTURE OF THE BATTERY SEAL AT THE TERMINAL AND THE PREMATURE FAILURE OF THE BATTERY.**

CAUTION **USE ONLY THE HARDWARE PROVIDED WITH THE BATTERY. DO NOT USE STAINLESS STEEL OR STEEL WASHERS BETWEEN THE RING TERMINAL AND THE BATTERY TERMINAL.**

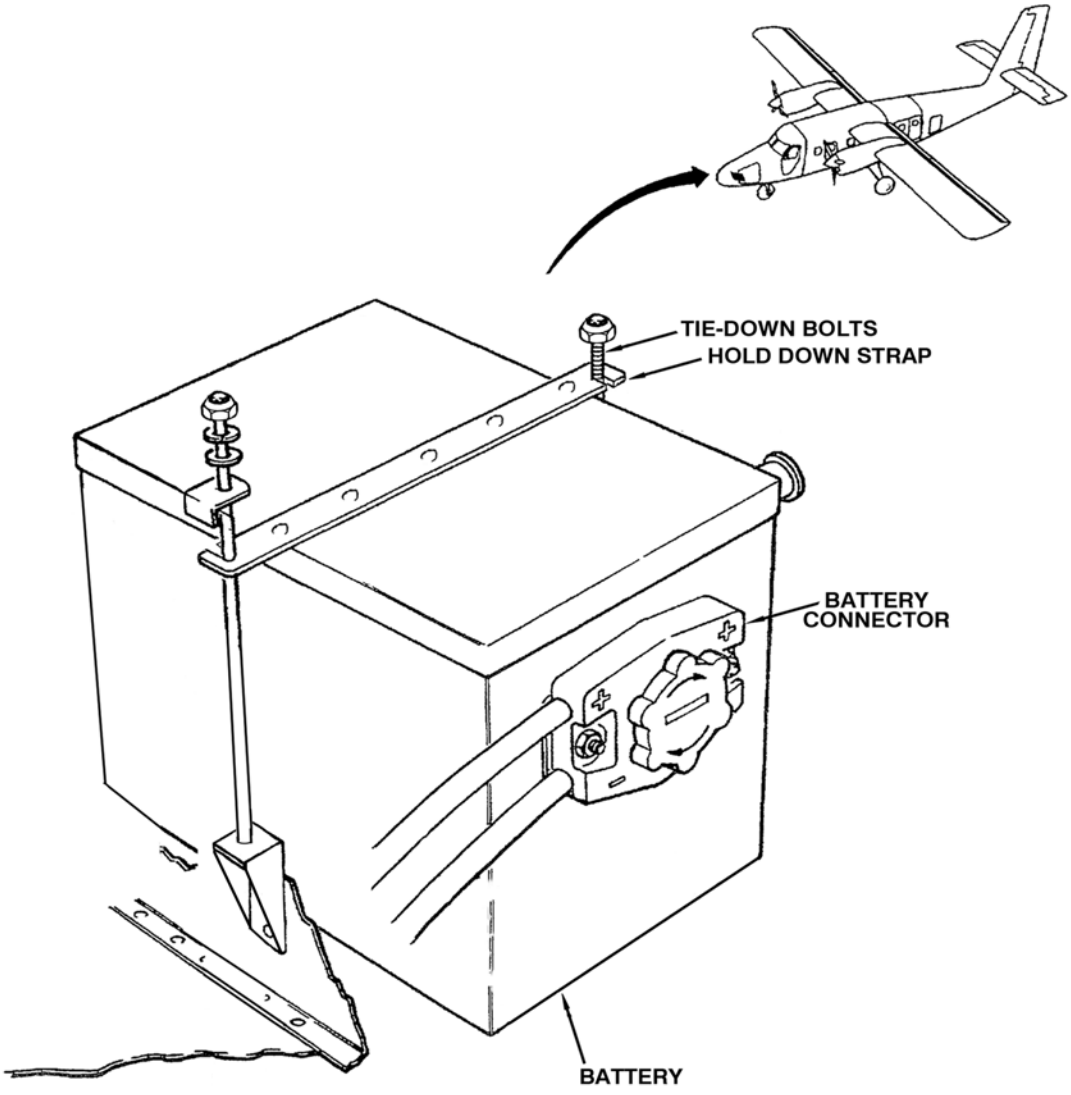
7) Engage battery hold down hardware, torque and lock wire battery tie down bolts. For aircraft equipped with S.O.O. 6371, torque both tie down bolts to snug. Then tighten each bolt, one additional turn (360 degrees).

8) Connect the battery quick connect.

9) Replace the access panel in the rear baggage compartment floor.



Main Battery Installation
Figure 1



Main Battery Installation (S.O.O. 6371)
Figure 2

2. Adjustment/Test

A. Constant Voltage Charging

In the DHC-6 Series 400 aircraft, valve regulated lead acid batteries are used, and the constant voltage charging method is employed, using the voltage-regulated generator charging circuit. The graph shown in PSM 1-6-6A, Manufacturer's Information, indicates what happens essentially during this type of charge.

A voltage produced by the generator permits current to flow to the battery. In a discharged battery, the maximum surge of current will be approximately 10 times the rated capacity. The high surge currents are due to the low internal resistance of the battery. As the battery approaches a charged state, the current falls off rapidly and the voltage rises. The recommended voltage setting for the regulator, is computed by multiplying the number of cells in the battery by a factor of 1.5 to 1.55.

B. Apparent Loss of Capacity

When apparent loss or temporary loss of capacity occurs, the battery capacity will be lower than the rated capacity. The lower capacity is normally indicated by an abnormal slow engine starting performance or if abnormally high charging current is required to maintain the battery at bus voltage.

C. Capacity Test

1) Special Tools

a) Advanced Power Products Beta D–50 Aircraft Battery Analyzer, P/N 4146 or 4159 or equal.

b) Advanced Power Products Alpha C–25 Battery Charger Part Number 4142 or 4159 or equal.

WARNING **CONTACT CONCORDE FOR DETERMINATION OF EQUAL TEST EQUIPMENT. SOME BRANDS OF BATTERY CHARGERS WILL DESTROY BATTERY.**

2) Procedure

a) Remove battery from aircraft as per Para 1.A.

b) Stabilize the battery at 15°C or higher. The battery must be at this temperature for at least 24 hours.

c) If battery is not fully charged at this point, charge it as per Para 3.B.

d) Discharge the battery at the C1 rate on the label to the end point voltage (EPV) of 20 volts.

e) Record the time to the end point voltage (EPV).

f) The battery is acceptable for continued use if the ampere hour capacity (actual hours if discharge X ampere rate of discharge) is greater than 85% of the normal capacity (C1) shown on the battery label (i.e. 51 minutes or more).

g) If the battery passes first capacity test, charge the battery as per Para 3.B and return to service.

h) If the battery failed the capacity test for the first time, follow the battery conditioning procedure as per Para 3.C.

i) If the battery failed the capacity test for a second time, replace the battery.

j) After battery has been conditioned, repeat the capacity test.

Note If the battery gets very hot (external case temperature greater than 55°C (130°F)) during charge, it should be replaced.

- If battery passes, refer to Para 2.C.2).g).
- If the battery fails, replace it.

k) Install battery as per Para 1.B.

3. Inspection/Check

A. Inspect Main Battery

A visual inspection in the aircraft should be made of the battery, and battery connector, receptacle and vent lines should be checked for contact failure or damage and the allen head screws or nuts should be torqued, as per manufacturer's data.

B. Charge the Battery

1) Special Tools

a) Advanced Power Products Beta D–50 Aircraft Battery Analyzer, P/N 4126, or equal.

b) Advanced Power Products Alpha C–25 Battery Charger, P/N 4142, or equal.

c) Advanced Power Products CA15–50 Charger/analyzer P/N 4159, or equal.

WARNING **CONTACT CONCORDE FOR DETERMINATION OF EQUAL TEST EQUIPMENT. SOME BRANDS OF BATTERY CHARGERS WILL DESTROY THE BATTERY.**

2) Depending on the type of charger available, charge the battery Constant Potential (CP). Charge at 28.2 volts for 24 volt batteries until the charge current stabilizes for 1 hour.

WARNING **ONLY CONSTANT POTENTIAL CHARGING MAY BE DONE ON THE AIRCRAFT. DO NOT CONSTANT CURRENT CHARGE A BATTERY ON THE AIRCRAFT. THERE MAY BE A SERIOUS RISK OF INJURY TO PERSONNEL AND/OR DAMAGE TO THE AIRCRAFT OR AIRCRAFT SYSTEMS.**

C. Battery Conditioning Procedure

WARNING **REMOVE BATTERY FROM AIRCRAFT PRIOR TO DOING A BATTERY CONDITIONING PROCEDURE.**

1) Special Tools

a) Advanced Power Products Beta D–50 Aircraft Battery Analyzer, P/N 4146 or 4159 or equal.

b) Advanced Power Products Alpha C–25 Battery Charger P/N 4142 or 4159 or equal.

WARNING **CONTACT CONCORDE FOR DETERMINATION OF EQUAL TEST EQUIPMENT. SOME BRANDS OF BATTERY CHARGERS WILL DESTROY BATTERY.**

2) Procedure

a) Remove battery from aircraft as per Para 1.A.

b) Discharge battery at C1 rate, noted on the battery label, to end point voltage of 18 volts.

c) Constant current charge at the (C1/10) rate for 16 hours or until the battery voltage on charge is 31 volts or higher for 4 hours.

Note C1 as shown on battery label.

WARNING **BATTERY CAN BE DAMAGED IF CONDITION PROCEDURE DONE TOO MANY TIMES.**

d) Allow battery to cool down. (Approximately 8 hours).

e) After battery has been conditioned, repeat capacity test as per Para 2.C.

4. Battery Top Charging using Ground Power (Mod 6/2314)

WARNING **ONLY CONSTANT POTENTIAL CHARGING MAY BE DONE ON THE AIRCRAFT. DO NOT CONSTANT CURRENT CHARGE A BATTERY ON THE AIRCRAFT, THERE MAY BE A SEROUS RISK OF INJURY TO PERSONNEL AND/OR DAMAGE TO THE AIRCRAFT OR AIRCRAFT SYSTEM.**

CAUTION **THIS PROCEDURE IS ACCEPTABLE ONLY WHEN A KNOWN GOOD BATTERY HAS BEEN DISCHARGED THROUGH EXTENDED GROUND USE AND THERE IS CONCERN ABOUT ITS CAPABILITY TO POWER THE AIRCRAFT AND AVIONICS DURING AN ENGINE START.**

CAUTION **THIS PROCEDURE REQUIRES A KNOWN GOOD EXTERNAL GROUND POWER SOURCE THAT IS CAPABLE OF PROVIDING CLEAN AND CONSTANT DC VOLTAGE AND HAS AN AMPERE RATING CAPABLE OF POWER THE BASIC AIRCRAFT (APPROXIMATELY 35 AMPERES) PLUS BATTERY CHARGING CURRENT WHICH CAN EXCEED 100 AMPERES (135 AMPERES TOTAL).**

When Mod 6/2314 is embodied, it is possible to turn on EXTERNAL power and BATTERY power at the same time. When the BATTERY and EXTERNAL switches are both ON it is possible to top charge the battery using ground power.

Note This procedure does not replace the battery conditioning procedure or the battery charge procedure.

A. Check the battery voltage:

1) BATTERY switch ON, EXTERNAL switch OFF.

2) When the cockpit displays illuminate, check BATTERY VOLTS and AMPS.

3) If the BATTERY VOLTS indicates 20 Volts or lower, do not use this procedure. Refer to Para 3.B.

4) Battery discharge current is displayed as a negative number, typically –35 A, depending on aircraft electrical loads.

B. Connect ground power:

1) With the EXTERNAL switch OFF, connect 28 Volt DC power source to external power receptacle.

2) Check the GND PWR voltage displayed on the avionics display.

3) Set the ground power voltage as shown in the following table. The voltage is displayed on the cockpit display.

CAUTION **BATTERY CAN BE OVERCHARGED BY SETTING THE GROUND POWER VOLTAGE TOO HIGH. HIGH CHARGING VOLTAGE WILL SHORTEN THE BATTERY LIFE SIGNIFICANTLY.**

Battery Temperature (use OAT reading)	Ground Power Voltage Setting (Volts DC)
Below 0°C (32°F)	29.0 – 29.5
0 to 15°C (32 to 59°F)	28.5 – 29.0
16 to 30°C (60 to 86°F)	28.0 – 28.5
31 to 45°C (87 to 113°F)	27.5 – 28.0
Above 45°C (113°F)	27.0 – 27.5

4) While observing the cockpit electrical display, switch the EXTERNAL switch ON. GND PWR turns green. BATTERY V and GND PWR V displays the same value (within 0.1 volt). A slight drop of GND PWR voltage is acceptable depending on the capability of the external DC power source.

5) Immediately check the battery current. The value on the cockpit display is positive indicating battery charge. (The cockpit dipslay is capable of presenting values between –100 A and + 100 A. A blank value indicates out of range, i.e. > 100 Amps).

WARNING **IF THE READING EXCEEDS THE CAPABILITY OF THE EXTERNAL DC POWER SOURCE, OR IF THE VOLTAGE IS FLUCTUATING, SET THE EXTERNAL SWITCH TO OFF IMMEDIATELY AND INVESTIGATE.**

Note Normal top charge is indicated by slowly decreasing battery charge current. Typical values start at below 70 Amps decreasing to 5 Amps, the top charge is complete.

6) When the charge current has dropped to less than 5 Amps, the top charge is complete.

7) Set BATTERY and EXTERNAL switches as required.